

2. Implement a basic finite state automaton that recognizes a specific language or pattern. In this example, we'll create a simple automaton to match strings ending with 'ab' using python.

1
0s

```
def ends_with_ab(string):  
    state = 0  
  
    for char in string:  
        if state == 0:  
            if char == 'a':  
                state = 1  
            else:  
                state = 0  
        elif state == 1:  
            if char == 'b':  
                state = 2  
            elif char == 'a':  
                state = 1  
            else:  
                state = 0  
        elif state == 2:  
            if char == 'a':  
                state = 1  
            else:  
                state = 0  
  
    return state == 2  
  
# Test  
words = ["ab", "aab", "babab", "abc", "ba"]  
for w in words:  
    print(w, "->", ends_with_ab(w))
```

```
print(w, "->", ends_with_ab(w))
```

```
... ab -> True  
    aab -> True  
    babab -> True  
    abc -> False  
    ba -> False
```